

The allegory axioms, and what defines them in the Frobenius calculus

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$F(x) \leq F(x) \implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x)))$$

$$G(y) \leq G(y) \implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y)$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

- write F as $/$, G as $\backslash$

- render $a \leq b$ as $\boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}}$

- **rewrite** $X \leq XF$ as $1 \leq F$

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\boxed{\frac{\backslash X}{Y}} = \boxed{\frac{X}{/Y}}$$

ladj

$$\begin{array}{c} \boxed{\frac{X/}{X/}} \xRightarrow{\{\text{adj}\}} \boxed{\frac{X}{X\backslash}} \xRightarrow{\{\text{rewrite}\}} \boxed{\frac{*}{\bigwedge}}_{\text{unit}} \xRightarrow[\{\text{put } / \text{ on right}\}]{\{\text{put } \backslash \text{ on left}\}} \boxed{\frac{\backslash}{\bigwedge}} \\ \boxed{\frac{Y\backslash}{Y\backslash}} \xRightarrow{\{\text{adj}\}} \boxed{\frac{Y\bigvee}{Y}} \xRightarrow{\{\text{rewrite}\}} \boxed{\frac{\bigvee}{*}}_{\text{counit}} \xRightarrow[\{\text{put } \backslash \text{ on right}\}]{\{\text{put } / \text{ on left}\}} \boxed{\frac{\bigvee}{/}} \end{array}$$

now we have $\bigvee = /$ and $\bigwedge = \backslash$, and you just discovered string diagrams.

§1.1. Every adjunction in Rel

Both halves of a row are written as sections — the operator stays and its missing argument is the gap: $\cdot S$ is $R \mapsto R \cdot S$, $/S$ is $T \mapsto T/S$, $S \setminus$ is $Y \mapsto S \setminus Y$. Composition is juxtaposition, so the dot is written only in the rows where nothing else marks the gap.

$F \dashv G$	F's type	$1 \sqsubseteq FG$	$GF \sqsubseteq 1$	F preserves u	G preserves n	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangleright$	$A \rightarrow A \otimes A$	$1 \sqsubseteq \triangleleft \triangleright$	$\triangleright \triangleleft \sqsubseteq 1$	$(RuS) \triangleleft =$ $R \triangleleft uS \triangleleft$	$(RnS) \triangleright =$ $R \triangleright nS \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
$\multimap \dashv \multimap$	$A \rightarrow I$	$1 \sqsubseteq \multimap \multimap$	$\multimap \multimap \sqsubseteq 1$	$(RuS) \multimap =$ $R \multimap uS \multimap$	$(RnS) \multimap =$ $R \multimap nS \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R = R \circ \circ$	$R = R \circ \circ$	$(RuS) \circ =$ $R \circ uS \circ$	$(RnS) \circ =$ $R \circ nS \circ$	$R \circ \circ \circ = R \circ$	$R \circ \circ \circ = R \circ$
$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \sqsubseteq R \cap R$	$R \cap S \sqsubseteq R$	—	—	$R \cap R = R$	$R \cap R = R$
$U \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$R \sqsubseteq RuS$	$RuR \sqsubseteq R$	—	—	$RuR = R$	$RuR = R$
$\perp \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$	—	$\perp \sqsubseteq R$	—	—	—	—
$\cdot S \dashv /S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	$R \sqsubseteq (RS)/S$	$(T/S)S \sqsubseteq T$	$(RuT)S =$ $RSuTS$	$(RnT)/S =$ $R/SnT/S$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$R \sqsubseteq S \setminus (SR)$	$S(S \setminus T) \sqsubseteq T$	$S(RuT) =$ $SRuST$	$S \setminus (RnT) =$ $S \setminus RnS \setminus T$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \sqsubseteq R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \sqsubseteq Y$	$R \cap (XuY) =$ $(R \cap X) \cup (R \cap Y)$	$R \Rightarrow (X \cap Y) =$ $(R \Rightarrow X) \cap (R \Rightarrow Y)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \sqsubseteq (\mathcal{D}R)T$	$\mathcal{D}(AT) \sqsubseteq A$	$\mathcal{D}(RuS) =$ $\mathcal{D}Ru\mathcal{D}S$	$(AnB)T =$ $AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \sqsubseteq T(\mathcal{R}R)$	$\mathcal{R}(TA) \sqsubseteq A$	$\mathcal{R}(RuS) =$ $\mathcal{R}Ru\mathcal{R}S$	$T(AnB) =$ $TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	$1 \sqsubseteq ff^\circ$	$f^\circ f \sqsubseteq 1$	$(RuS)f =$ $RfuSf$	$(RnS)f^\circ =$ $Rf^\circ nSf^\circ$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$f^\circ \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	$1 \sqsubseteq ff^\circ$	$f^\circ f \sqsubseteq 1$	$f^\circ (XuY) =$ $f^\circ Xu f^\circ Y$	$f(X \cap Y) =$ $fX \cap fY$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$i \dashv P$	$\text{Map} \hookrightarrow \text{Rel}$	$\{\cdot\} : A \rightarrow P \ A$	$\exists : P \ B \rightarrow B$	—	—	$\Lambda(\exists) = 1$	$\Lambda(R)\exists = R$

$\cdot S : B \rightarrow C$ in the $\cdot S$ row and $S \cdot : A \rightarrow B$ in the $S \cdot$ one; f is a map, $B \rightarrow C$ in the $\cdot f$ row and $A \rightarrow B$ in the $f^\circ \cdot$ one. $\text{Cor } A$ is the poset of coreflexives on A : the A and B of the $\mathcal{D}$ row live in $\text{Cor } A$, those of the $\mathcal{R}$ row in $\text{Cor } B$ — coreflexives, not objects.

§1.2. Fixed points

$F \dashv G$	closed: $XFG=X$	open: $YGF=Y$
$\triangleleft \dashv \triangleright$	$X \triangleleft \triangleright = X$	$Y \triangleright \triangleleft = Y$
$\multimap \dashv \multimap$	$X \multimap \multimap = X$	$Y \multimap \multimap = Y$
$\circ \dashv \circ$	every X	every Y
$\Delta \dashv \cap$	every X	$Y_1 = Y_2$
$U \dashv \Delta$	$X_1 = X_2$	every Y
$\perp \dashv !$	—	$Y = \perp$
$\cdot S \dashv /S$	$(XS)/S = X$	$(Y/S)S = Y$
$S \cdot \dashv S \setminus$	$S \setminus (SX) = X$	$S(S \setminus Y) = Y$
$R \cap \dashv R \Rightarrow$	$R \Rightarrow (X \cap R) = X$	$R \cap (R \Rightarrow Y) = Y$
$\mathcal{D} \dashv \cdot T$	$(\mathcal{D}X)T = X$	$\mathcal{D}(AT) = A$
$\mathcal{R} \dashv T \cdot$	$T(\mathcal{R}X) = X$	$\mathcal{R}(TA) = A$
$\cdot f \dashv \cdot f^\circ$	$Xff^\circ = X$	$Yf^\circ f = Y$
$f^\circ \cdot \dashv f \cdot$	$ff^\circ X = X$	$f^\circ fY = Y$
$i \dashv P$	—	—

F and G restrict to mutually inverse isomorphisms between the two columns.

§1.3. §2.314, and what composing rows gives

One more adjunction first, which §1.1 cannot hold because it is ANTITONE — its left adjoint turns joins into meets, so that table's F preserves u column would read the wrong way:

$F \dashv G$	the law	both units	Lean
$R/ \dashv \backslash R$	$T \sqsubseteq R/S$ iff $S \sqsubseteq T \backslash R$	$S \sqsubseteq (R/S) \backslash R$, $T \sqsubseteq R/(T \backslash R)$	<code>le_div_iff</code> then <code>le_leftDiv_iff</code>

Two rows of §1.1 composed are a third adjunction, and its right adjoint read in the two orders is a law. Row first, column second; right adjoints compose the other way round, which is where every reversal in the table comes from.

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	$T n \cdot$	$^\circ$	Δ
$\cdot S$	$R/(ST) = (R/T)/S$ <code>div_comp_assoc</code>	$T \backslash (R/S) = (T \backslash R)/S$ <code>leftDiv_div</code>	$R/(Sg) = (Rg^\circ)/S$	$g(R/S) = (gR)/S$ <code>map_comp_div</code>	—	$(Y/S)^\circ = S^\circ \backslash (Y^\circ)$ <code>leftDiv_div_recip</code>	$(T_1 n T_2)/S = T_1/S n T_2/S$ <code>div_inter_eq</code>
$S \cdot$	$S \backslash (R/T) = (S \backslash R)/T$ <code>leftDiv_div</code>	$(TS) \backslash R = S \backslash (T \backslash R)$ <code>leftDiv_comp</code>	$S \backslash (Rg^\circ) = (S \backslash R)g^\circ$	$(g^\circ S) \backslash R = S \backslash (gR)$	—	$(S \backslash Y)^\circ = Y^\circ/S^\circ$ <code>leftDiv_div_recip</code>	$S \backslash (T_1 n T_2) = S \backslash T_1 n S \backslash T_2$ <code>leftDiv_inter</code>
$\cdot f$	$R/(fT) = (R/T)f^\circ$ <code>div_comp_recip_map</code>	$T \backslash (Rf^\circ) = (T \backslash R)f^\circ$	$(fg)^\circ = g^\circ f^\circ$ <code>recip_comp</code>	—	—	$(fY)^\circ = Y^\circ f^\circ$ <code>recip_comp</code>	$(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$
$f^\circ \cdot$	$f(R/T) = (fR)/T$ <code>map_comp_div</code>	$(Tf^\circ) \backslash R = f(T \backslash R)$	—	$(fg)^\circ = g^\circ f^\circ$ <code>recip_comp</code>	—	$(fY)^\circ = Y^\circ f^\circ$ <code>recip_comp</code>	$f(T_1 n T_2) = fT_1 n fT_2$
$R n \cdot$	—	—	—	—	$(R n T) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	$R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$
$^\circ$	$(Y/T)^\circ = T^\circ \backslash (Y^\circ)$ <code>leftDiv_div_recip</code>	$(T \backslash Y)^\circ = Y^\circ/T^\circ$ <code>leftDiv_div_recip</code>	$(gY)^\circ = Y^\circ g^\circ$ <code>recip_comp</code>	$(gY)^\circ = Y^\circ g^\circ$ <code>recip_comp</code>	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$Y^\circ = Y$ <code>recip_recip</code>	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$ <code>recip_inter</code>
u	$(X_1 u X_2)T = X_1 T u X_2 T$ <code>union_comp_distrib</code>	$T(X_1 u X_2) = T X_1 u T X_2$ <code>comp_union_distrib</code>	$(X_1 u X_2)g = X_1 g u X_2 g$ <code>union_comp_distrib</code>	$g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$ <code>comp_union_distrib</code>	$T n (X_1 u X_2) = (T n X_1) u (T n X_2)$	$(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$ <code>recip_union</code>	—
$\perp$	$\perp T = \perp$ <code>zero_comp</code>	$T \perp = \perp$ <code>comp_zero</code>	$\perp g = \perp$ <code>zero_comp</code>	$g^\circ \perp = \perp$ <code>comp_zero</code>	$T n \perp = \perp$	$\perp^\circ = \perp$	—

Every pair composes — the parameters can always be chosen so the hom-sets line up — so — never means a type clash: it means the equation the pair yields is associativity, or has no name. The table reads the same across the diagonal: a cell and its mirror are one composite read in the two orders, so they carry one law with the two parameters swapped. The $^\circ$ row and column are that law's own mirror, which is why they turn every $/$ into a $\backslash$.

A cell names the Lean theorem that IS it, or the one it is an instance of: the four $^\circ$ division cells are `leftDiv_div_recip` at $S=1$ or $T=1$, and the `recip_comp` cells are the allegory axiom itself, as is `recip_recip`. A cell with NO name is derived here from uniqueness of right adjoints and is not in the repository — the six map-and- $\backslash$ cells and both Heyting cells. `heytingImpl` in `Freyd/S2_30.lean` is not those: it is $A \supset B \triangleq 1 \wedge B/A$ on (a, a) alone, while this row's $\Rightarrow$ runs over a whole hom-set.

$\Delta \dashv n$ is a column and $u \dashv \Delta$, $\perp \dashv !$ are rows, each on one side only: they run between a hom-poset and its square, or the one-point poset, and nothing else starts at a square or ends at the one-point poset, so the other side would be an empty line. What those three give back is §1.1's own columns. $\Delta \dashv n$ after a row $F \dashv G$ composes to $X \mapsto (F X, F X)$, which is also Δ followed by F on each coordinate; the two right adjoints must agree, and that is $G(T_1 n T_2) = G T_1 \wedge G T_2$ — the G preserves n column. $u \dashv \Delta$ before a row makes the two right adjoints agree outright, so the LEFT ones do: $F(X_1 u X_2) = F X_1 \cup F X_2$. $\perp \dashv !$ is the same with no coordinates at all, giving $F \perp = \perp$, which §1.1 carries no column for. Their three mutual cells are — because there the law is componentwise and says nothing. Lean writes the bottom $\emptyset$, not $\perp$.

Everything below is a row of §1.1 read at a chosen argument.

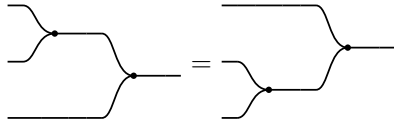
what you get	which move	Lean
$(R/S)(S/W) \sqsubseteq R/W$ [2.314]	count of $\cdot W \dashv /W$ at S , then count of $\cdot S \dashv /S$ at R	<code>div_comp</code>
$1 \sqsubseteq R/R$ [2.314]	unit of $\cdot R \dashv /R$ at 1	<code>one_le_div_self</code>
$(R/R)(R/R) = R/R$ [2.314]	$\sqsubseteq$ is the first line at $S=W=R$, $\supseteq$ the second	<code>div_self_comp_self</code>
$(R/R)R = R$ [2.314]	$\sqsubseteq$ is the counit at R , $\supseteq$ is the second line composed with R	<code>div_self_comp</code>
$R/1 = R$ [2.314]	$\cdot 1 \dashv /1$ is the identity adjunction	<code>div_one</code>
$R/(SuT) = R/S n R/T$ [2.314]	the antitone row above: a join in the divisor turns into a meet	<code>div_union</code>

Composing is `Freyd.Adj.comp`, uniqueness is `Freyd.Adj.right_unique`, both in `Freyd/S2_313.lean`. `postDiv_comp` is the second name the cross table's $\cdot S/\cdot T$ cell goes by.

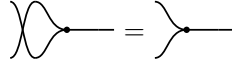
§2. Relations

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, {}^\circ)$ where $\otimes$ is commutative cartesian product, and converse ${}^\circ \dashv$, and

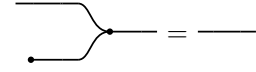
$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ a commutative monoid with $\mathbf{C}^\circ \dashv \mathbf{C}$. We write $\mathbf{C}^\circ$ as $(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$



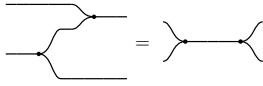
$\triangleright$ associative



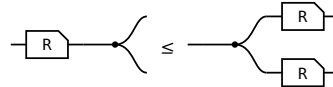
$\triangleright$ commutative



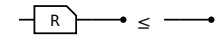
$\multimap$ is its unit



Frobenius, one half — the other is its ${}^\circ$



$R\triangleleft \leq \triangleleft(R\otimes R)$

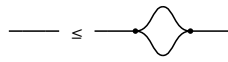


$R\multimap \leq \multimap$

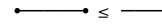
§2.1. $\forall$ object $\mathbf{A}$, $(\mathbf{A}, \triangleleft, \multimap) \dashv (\mathbf{A}, \triangleright, \multimap)$



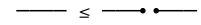
$\triangleright \triangleleft \leq \mathbb{1}$



$\mathbb{1} \leq \triangleleft \triangleright$



$\multimap \multimap \leq \mathbb{1}$

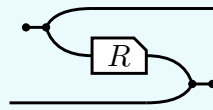


$\mathbb{1} \leq \multimap \multimap$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

$^\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



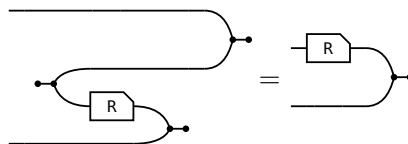
$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\circlearrowright$ are $\triangleright^\circ$ and $\circlearrowleft^\circ$. The laws of $^\circ$ are that it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

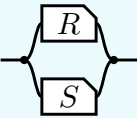
The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circlearrowright$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circlearrowleft \triangleleft$ bends the a strand down and the $\triangleright \circlearrowright$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

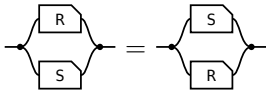
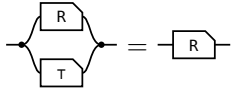
§4. $\cap$ is a commutative idempotent monoid on every hom-set

The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



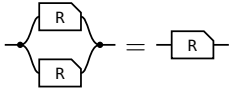
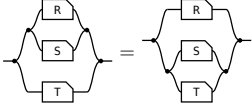
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).



unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$

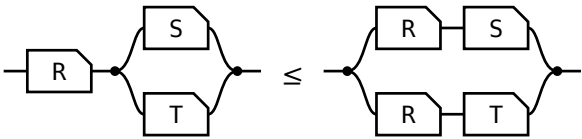
commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F (R \cap S) = F R \cap F S$.	

§5. Union

Past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure. It draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. The residual, which needs a second composition with linear adjoints, is in the division section below.

statement	picture
$R \sqsubseteq R \cup S$ union $R \ S := \langle R, S \rangle [1, 1]$: offer both branches, then forget which was taken.	
$R \cup S = S \cup R$	
$\perp \cup R = R$ $\perp$ routes through the zero object; there is no shape for it.	
$R (S \cup T) = R S \cup R T$ — composition distributes over union, which $\cap$ does not.	
$(R \cup S)^\circ = R^\circ \cup S^\circ$	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole distributive , and the one picture with both layers in it.	

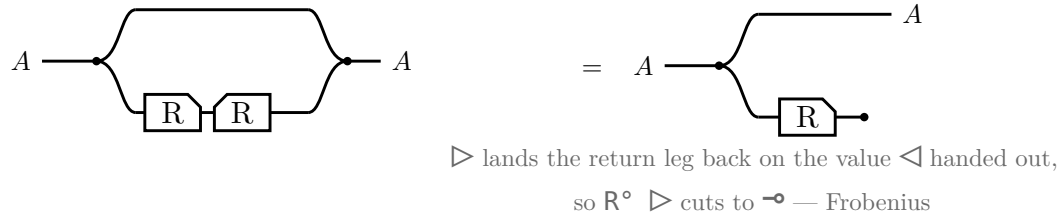
The last of Freyd's §2.21 equations that have no picture here — a tape around nothing, or a tape around what is already inside it, is not worth drawing. R = people who admire a , S = people who admire b throughout.

equation	the reading
$R \cup (S \cap R) = R$ $(R \cup S) \cap R = R$ absorption	People who admire a , or who admire both: still the people who admire a . People who admire a or b , and who admire a : the same. The smaller side of each pair is already inside the larger.

equation	the reading
$R \perp = \perp R$ $0_S = 0_{\{R S\}}$ — and $\perp R = \perp$ on the other side	Admiring someone and then taking a step that leads nowhere gets you nowhere. $\perp$ is a two-sided zero for composition, as τ is not.

§6. Domain and range

The **domain** $\text{Dom } R \triangleq 1 \cap R R^\circ$ and the **range** $\text{Ran } R \triangleq \text{Dom } R^\circ$.

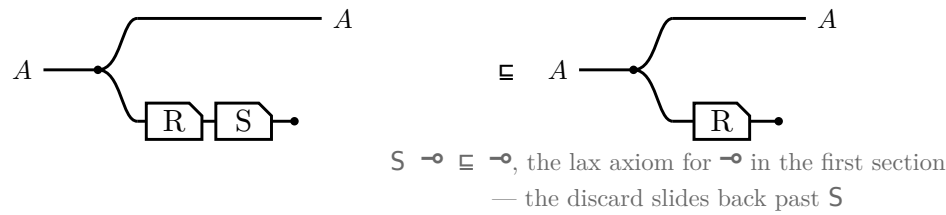


Running R and throwing the result away leaves only the fact that R could fire, and $\text{Ran } R$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

$\text{Dom } R \triangleq 1 \cap R R^\circ$
$\text{Dom } R \subseteq A \iff R \subseteq A R, \text{ for } A \text{ coreflexive}$
$\text{Dom } (R S) \subseteq \text{Dom } R$
$\text{Dom } (R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Dom } R = 1 \iff 1 \subseteq R R^\circ$
$R \text{ simple} \iff R^\circ R \subseteq 1$
$R \text{ a map} \iff R \text{ entire and simple}$
$R, S \text{ entire} \implies R S \text{ entire} \text{ — likewise simple, likewise maps}$
$R S \text{ entire} \implies R \text{ entire}$

§6.1. Sliding the discard

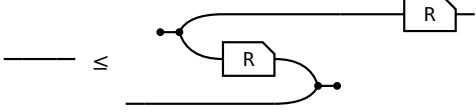
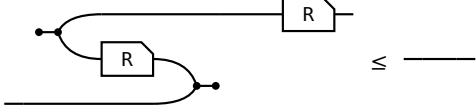
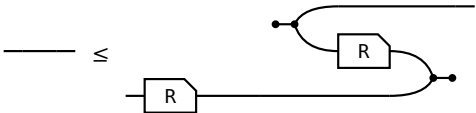
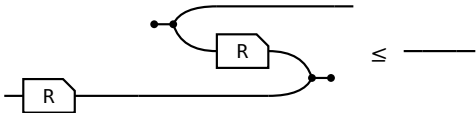
$\text{Dom } (R S) \subseteq \text{Dom } R$, and a single glyph for Dom would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

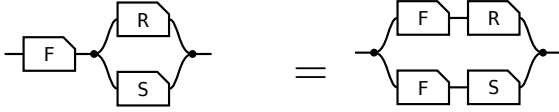


Equality is $S \text{ entire}$, which is the same picture read as $\text{Dom } R = 1 \iff R \text{ entire}$.

§7. Maps

Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by taking $\circ$ of those two. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

 surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	 single valued $R^\circ R \sqsubseteq 1$
 entire $1 \sqsubseteq R R^\circ$. With single valued, a map.	 injective $R R^\circ \sqsubseteq 1$

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = people x may ask, R = admires a, S = admires b. Single valued means one person, so the sides agree.	 one person who admires both a at A, b at B

§8. / is all of

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

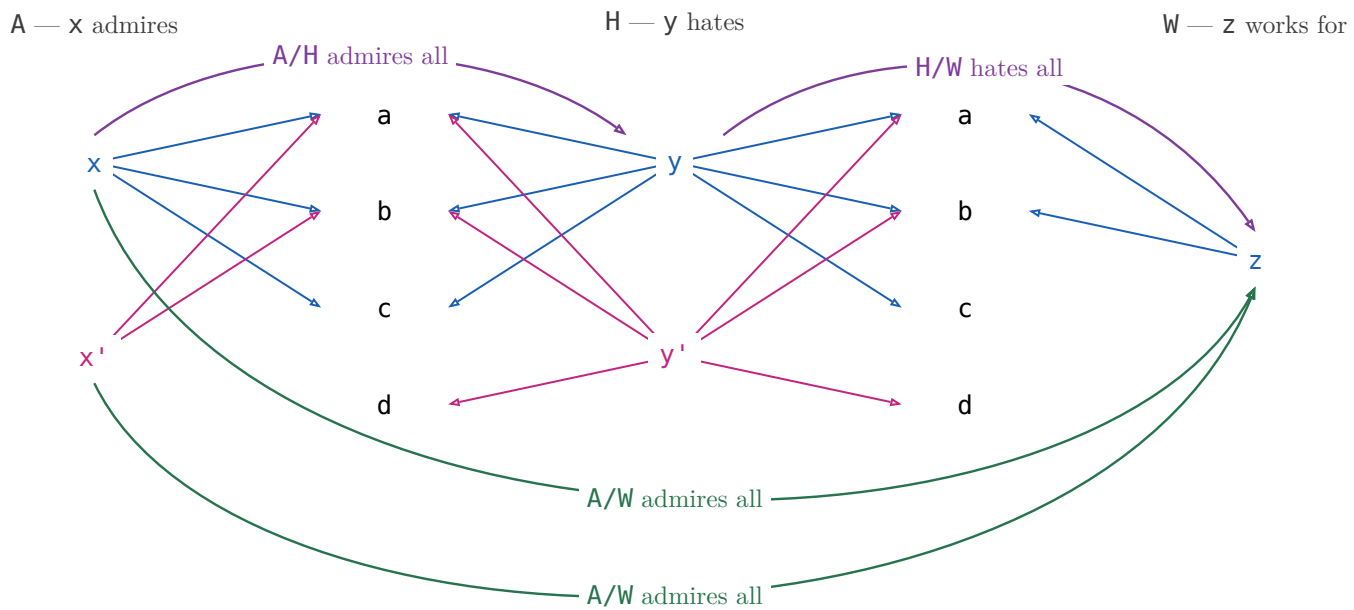
$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/ compares images: $S(y) \subseteq R(x)$. $\setminus$ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§8.1. $(R/S)(S/W) \sqsubseteq R/W$ 

$x (A/H) y$ — x admires everyone y hates

$y (H/W) z$ — y hates everyone z works for

$x (A/W) z$ — x admires everyone z works for

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \sqsubseteq R/S \iff X S \sqsubseteq R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \sqsubseteq S \setminus R \iff S X \sqsubseteq R$ The mirror — divide on the left when x comes first.	
$(R/S) S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S = \emptyset$: R/S is everyone, $(R/S) S = \emptyset$.	
$S (S \setminus R) \sqsubseteq R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	

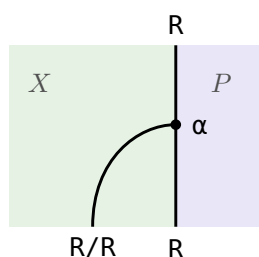
maps: $f (R/S) = (f R)/S$ Rename x before or after dividing — the licence to write $f R / S$.	
$R/(f S) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S)(S/W) \sqsubseteq R/W$ Someone who admires all of a hate-list that already covers everyone z works for admires those people too.	
$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R)(R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
$(R/R) R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's list is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-list is admiring each list in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: n , u , $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§8.2. R/R is a preorder, which is what a monad is here

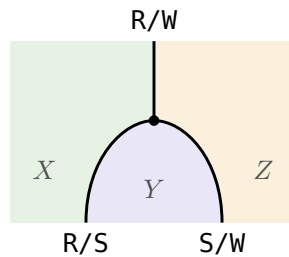
A **monad** in a locally posetal 2-category is a 1-cell M with $1 \sqsubseteq M$ and $M M \sqsubseteq M$ — a unit and a multiplication, and with the hom-posets thin there is nothing else to give. In $\mathbf{Rel}$ that is a reflexive transitive relation: a preorder. The table above proves both halves at $M := R/R$, and names the preorder they define.

$(R/R) R = R$ is then the action that makes R a module over that monad — Hinze and Marsden's $\alpha : M \circ A \rightarrow A$ (IntroString p. 83, laws (3.10a) and (3.10b)) at $M := R/R$, $A := R$. Their two coherence conditions are free here: parallel 2-cells are unique, so any two of them are equal.



Green is X , where x lives; purple is P , where the p of the definition lives. $R/R : X \rightarrow X$ runs inside X and ends on $R : X \rightarrow P$ at the action α , which is $(R/R) R \sqsubseteq R$.

§8.3. The whole family is one preorder



The same dot with one region more: $R/S : X \rightarrow Y$ and $S/W : Y \rightarrow Z$ come in, $R/W : X \rightarrow Z$ goes out. Green X , purple Y , amber Z .

Closing Y off is the mid-point y being quantified away; the $\sqsubseteq$ rather than $=$ is the loss the figure for that law above already draws.

Concretely in $\mathbf{Rel}$, $\mathbb{1} \sqsubseteq R/R$ and $(R/S)(S/W) \sqsubseteq R/W$ are one preorder on the disjoint union of $X, Y, Z, \dots$: reflexivity is $\mathbb{1} \sqsubseteq R/R$ inside a block, transitivity is $(R/S)(S/W) \sqsubseteq R/W$ across blocks.

Abstractly the same two are a **lax functor** out of the codiscrete category on $\{R, S, W, \dots\}$ — one object per relation, exactly one arrow each way — sending R to X , the object its rows are indexed by, and the arrow $R \rightarrow S$ to $R/S : X \rightarrow Y$. That is Bénabou's **polyad** (Bénabou 1967, Def. 5.5), of which a monad is the one-object case: the subsection above is this one at $R = S = W$.

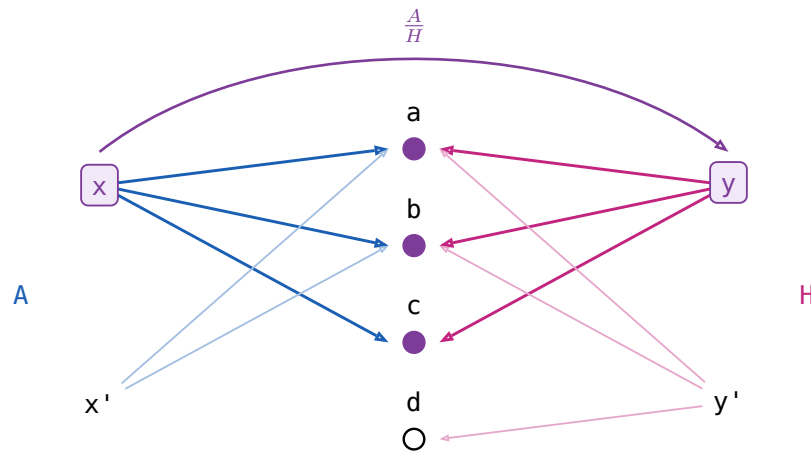
R/S is at the same time a bimodule between the two monads R/R and S/S : $(R/R)(R/S) \sqsubseteq R/S$ and $(R/S)(S/S) \sqsubseteq R/S$, each the counit $(R/S) S \sqsubseteq R$ with one action whiskered on — $(R/R) R = R$ on the left, $(S/S) S = S$ on the right.

§9. $x \text{ (Admires\%Hates) } y$ is x admires only all whom y hates

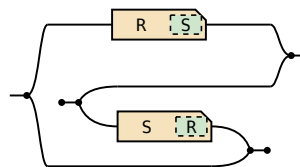
$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

$x \text{ (A/H) } y$ — x admires everyone y hates
 $x \text{ ((H/A)^\circ) } y$ — x admires only people y hates
 $x \text{ ((A/H) \cap (H/A)^\circ) } y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.



A meet of two long divisions, the second turned round by the converse frame:



exported from the definition, not transcribed

$\left(\frac{R}{S}\right)^\circ = \frac{S}{R}$ Matching is symmetric.	
$\frac{R}{S} \frac{S}{W} \sqsubseteq \frac{R}{W}$ And transitive.	

No pictures for the rest of §2.35: symmetric division is not built from the generators.

law	the reading
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R$ and $X^\circ R \sqsubseteq S$	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$\frac{R}{S} S \sqsubseteq R$	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$
$\frac{R}{R} R = R$	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y = x$ always qualifies ($1 \sqsubseteq R\%R$ below)
$1 \sqsubseteq \frac{R}{R}$	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$\left(\frac{R}{R}\right)^2 = \frac{R}{R}$	So the relation admires the same people is an equivalence relation.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric X	The largest symmetric arrow that leaves R alone.

law	the reading
$\frac{R}{1}$ is the simple part of R	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{Dom } \frac{R}{S} = 1 \cap (R/S) (S/R)$	Its domain is the domain of simplicity of R .

§9.1. Straight

S is **straight** when $\frac{S}{S} = 1$ — no two y s hate the same people.

law	the reading
every symmetric X with $X \circ S \sqsubseteq S$ is coreflexive	Equivalently, S is straight.
$f \circ S = g \circ S \implies f = g$	A straight S tells its y s apart, so it cancels on the right.
$S \circ R \text{ straight} \implies S \text{ straight}$	If the longer chain tells them apart, the first step already does.
$S \text{ straight} \iff R/S \text{ simple for all } R$	If no two y s agree, at most one x can match a given y .
$R = h \circ S$, h a cover, S straight	In an effective division allegory every arrow factors that way.

§10. Power allegories

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**off, subject to

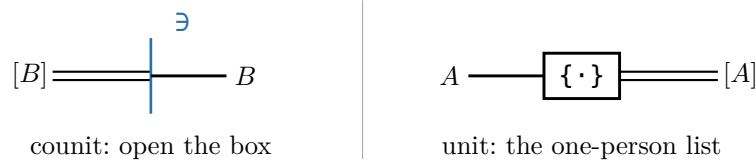
$$\begin{aligned} \exists_R \square &= R \square, & \exists_R &= \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R &\text{ is **thick** } \\ (\exists_R/\exists_R) \cap (\exists_R/\exists_R)^\circ &\subseteq \mathbb{1} & \exists_R &\text{ is **straight** } \end{aligned}$$

$R \square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : [B] \rightarrow B$, dropping the subscript.

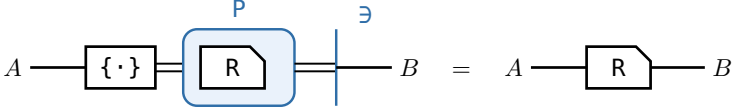
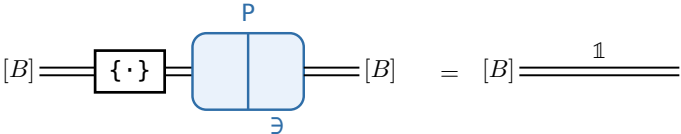
In **Rel** its source is the powerset of its target and $\mathfrak{l} \exists \mathfrak{i}$ iff $\mathfrak{i} \in \mathfrak{l}$, and $\exists$ reads a list $\mathfrak{l}$ back into the people on it.

$\Lambda(R) \triangleq \frac{R}{\exists} : A \rightarrow [B]$, for $R : A \rightarrow B$. The list-of map: send x to the list of the people x admires.

One convention: a $[B]$ -wire is drawn double, because it is a B -wire one level down — inside the box. The box is the power relator P , which the relator section below defines: a box holding $R : B \rightarrow C$ is $P R : [B] \rightarrow [C]$. Inside the box lives the allegory, outside only maps, and $\mathfrak{i} : \mathbf{Map}(\mathcal{A}) \hookrightarrow \mathcal{A}$ is the inclusion that forgets the difference. The adjunction $\mathfrak{i} \dashv P$, that is $\mathcal{A}(A, B) \cong \mathbf{Map}(A, [B])$, has the two ends:



Λ is neither of them — it is the transpose the adjunction gives, $\Lambda(R) = \{\cdot\} (P R)$: make the singleton, then push R through the box.

law	the reading
$\exists_R \square = R \square, \exists_R = \exists_R \square$	$\exists$ has the same target as R , and replacing R by the identity at that target leaves it unchanged: one $\exists$ per object, not per arrow.
$\exists$ is thick	Comprehension : every x has a list of exactly the people x admires. Equivalently every R factors as a map followed by $\exists$.
$\exists$ is straight , that is $\frac{\exists}{\exists} \subseteq \mathbb{1}$	Extensionality : two lists with the same people are the same list.
$\Lambda(R)$ is simple	$\exists$ is straight, and dividing by a straight arrow is simple. At most one list per x .
$\Lambda(R)$ is entire $\iff \exists$ thick	$\text{Dom } \frac{R}{\exists} = \mathbb{1} \cap (R/\exists)(\exists/R)$, the domain row above. At least one list per x , so $\Lambda(R)$ is a map .
$\Lambda(R) \exists = R$ Everything left of the $\exists$ wall is $\Lambda(R)$; open the wall and what was inside comes out unchanged. That is the β of this calculus.	
$\Lambda(R)$ is the only map with $\Lambda(R) \exists = R$	Two maps naming the same people name the same list — extensionality again.
$F \subseteq \Lambda(F \exists)$, F simple	A partial choice of lists is inside the total one.
$[\alpha] = \text{source of } \exists$, the power-object	Every list over α .
$\{\cdot\} \triangleq \Lambda(\mathbb{1})$, the singleton map , monic	The one-person list. $\Lambda(\mathbb{1})\Lambda^\circ(\mathbb{1}) \subseteq \frac{\mathbb{1}}{\exists} \subseteq \frac{\mathbb{1}}{\mathbb{1}} \subseteq \mathbb{1}$.
$\Lambda(\exists) = \mathbb{1}$ Make the list of a list, then read it back one level down. Straightness itself, and one of the two triangle identities of $\mathfrak{i} \dashv P$; the other is $\{\cdot\} \exists = \mathbb{1}$.	

law	the reading
fusion: $\Lambda(f \ R) = f \ \Lambda(R)$, f a map f is a plain rectangle — a map, so it may cross the $\{\cdot\}$ node. That is naturality of the unit, $f \ \{\cdot\} = \{\cdot\} \ (P \ f)$, and it is the whole content of fusion; a chamfered box is stuck on the side it is on.	
$\Lambda(f) = f \ \{\cdot\}$, f a map	Rename first or take singletons first — the fusion row above at $R = 1$.
$\frac{R}{S} = \Lambda(R) \Lambda^\circ(S)$ x and y match exactly when the two boxes hold one and the same list — the $[B]$ -wire joining them is the whole statement.	
$\mathbf{C}$ a topos $\implies \mathbf{Rel}(\mathbf{C})$ a power allegory	And back: a unitary tabular power allegory has $\mathbf{Map}(\mathbf{A})$ a topos. Extensionality and comprehension are all a topos adds.

§11. Relator

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S is $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F \mathbb{1} = \mathbb{1} \quad F(R \circ S) = (F R) (F S) \quad R \sqsubseteq S \implies F R \sqsubseteq F S$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

- For f a map, $F f$ is a map and $F(f^\circ) = (F f)^\circ$.
- Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$.
- A relator is fixed by what it does to maps.
- $F(R \cap S) \sqsubseteq (F R) \cap (F S)$, and strictly.
- $F(X \cap Y) = (F X) \cap (F Y)$ for X, Y coreflexive.
- $F(\text{Dom } R) = \text{Dom}(F R)$ for F preserving $\circ$.

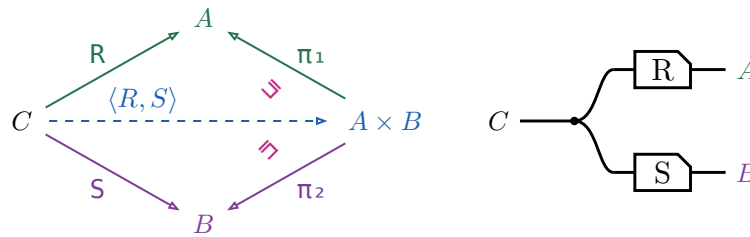
The **power relator** $P \multimap x (P R) y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P R \cap P S$, while $R \cap S = \emptyset$.

§11.1. The pairing $\langle R, S \rangle$

The **pairing** of $R : C \rightarrow A$ and $S : C \rightarrow B$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

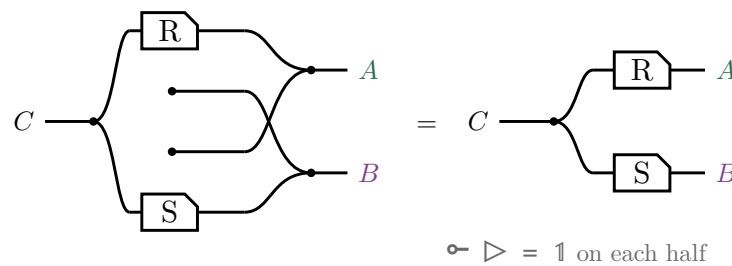
$$\langle R, S \rangle \pi_1 = (\text{Dom } S) R$$

$$\langle R, S \rangle \pi_2 = (\text{Dom } R) S$$



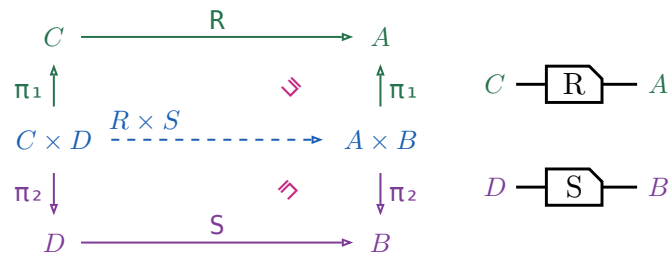
A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In Rel , $c \langle R, S \rangle (a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

No $\circ$ survives the translation. $\pi_1 = \mathbb{1} \otimes \multimap$ discards the second component, so $\pi_1^\circ = \mathbb{1} \otimes \multimap^\circ$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



§11.2. The relational product $R \times S$

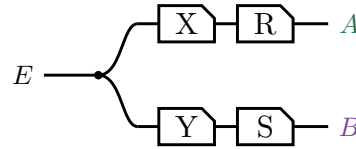
$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.



Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§11.3. Absorption

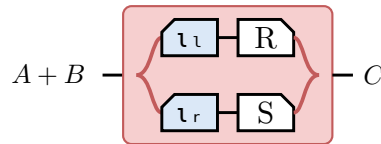
For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle X R, Y S \rangle$. Both sides are this picture:



§11.4. The coproduct $[R, S]$

The injections $\iota_l : A \rightarrow A + B$ and $\iota_r : B \rightarrow A + B$ are maps, and the coproduct they make of the maps stays a coproduct once every arrow is allowed: both equations hold with equality and $[R, S]$ is the only arrow satisfying them, with none of the Dom slack $\langle R, S \rangle$ carries.

$$[R, S] \triangleq \iota_l^\circ R \cup \iota_r^\circ S, \text{ and } R + S \triangleq [R \iota_l, S \iota_r].$$



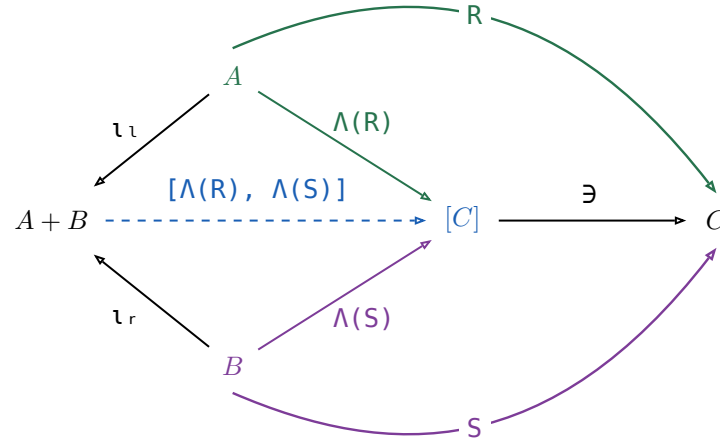
The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.

$\iota_l [R, S] = R, \iota_r [R, S] = S$, and $[R, S]$ is the only such arrow
$\iota_l \iota_l^\circ = 1 = \iota_r \iota_r^\circ$
$\iota_l \iota_r^\circ = \perp = \iota_r \iota_l^\circ$
$\iota_l^\circ \iota_l \cup \iota_r^\circ \iota_r = 1$
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$

The first row is not free: ι_l, ι_r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because Λ sends an arrow $A \rightarrow C$ to a map $A \rightarrow [C]$ reversibly, so the map coproduct can be applied underneath it. For any $T : A + B \rightarrow C$,

$$\begin{aligned}
 & \Leftrightarrow \iota_l T = R \text{ and } \iota_r T = S \\
 & \Leftrightarrow \Lambda \text{ is injective — } \Lambda(X) \ni = X \text{ gives } X \text{ back} \\
 & \Leftrightarrow \Lambda(\iota_l T) = \Lambda(R) \text{ and } \Lambda(\iota_r T) = \Lambda(S) \\
 & \Leftrightarrow \Lambda \text{ fusion, } \Lambda(f X) = f \Lambda(X) \text{ for } f \text{ a map} \\
 & \Leftrightarrow \iota_l \Lambda(T) = \Lambda(R) \text{ and } \iota_r \Lambda(T) = \Lambda(S) \\
 & \Leftrightarrow \text{the coproduct of maps, at the map } \Lambda(T) \\
 & \Leftrightarrow \Lambda(T) = [\Lambda(R), \Lambda(S)] \\
 & \Leftrightarrow \Lambda(X) \text{ is the only map with } \Lambda(X) \ni = X \\
 & \Leftrightarrow T = [\Lambda(R), \Lambda(S)] \ni
 \end{aligned}$$

Every step is an $\iff$, so $[\Lambda(R), \Lambda(S)] \ni$ is the **only** arrow satisfying the two equations — which is what $[R, S]$ was claimed to be. Drawn:



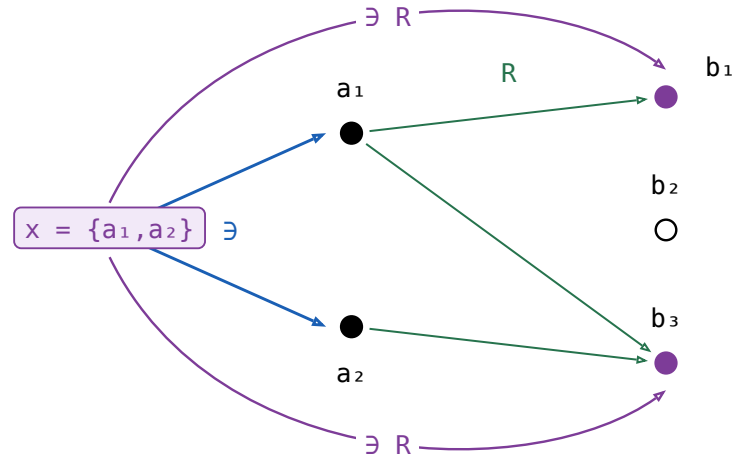
Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the pairing above. The border spells $[R, S] = [\Lambda(R), \Lambda(S)] \ni$, and pushing $\ni$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\iota_l^\circ \Lambda(R) \cup \iota_r^\circ \Lambda(S)) \ni = \iota_l^\circ R \cup \iota_r^\circ S$.

§11.5. The power relator $\mathbf{P} R$

For $R : A \rightarrow B$, $\mathbf{P} R \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : [A] \rightarrow [B]$.

$\exists R$ is a **composition** — $\exists : [A] \rightarrow A$ followed by R — and not Freyd's subscripted $\exists_R$, which is the $\exists$ at R 's **target**: §11 writes one $\exists$ per object and drops the subscript. Piece by piece, on one instance, with x , y lists and a , b their elements.

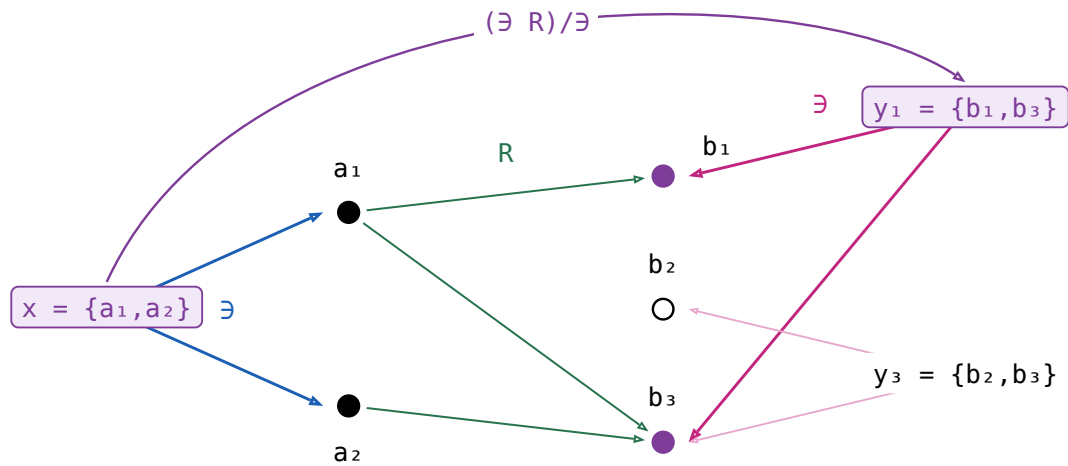
$x = \{a_1, a_2\}$, and R sends a_1 to b_1 and b_3 , a_2 to b_3 alone. Nothing on x reaches b_2 , so b_2 is drawn hollow.



$$\exists R : [A] \rightarrow B \quad x (\exists R) b \iff \exists a \in x. a R b$$

one arc per element of B that x reaches, and b_2 gets none

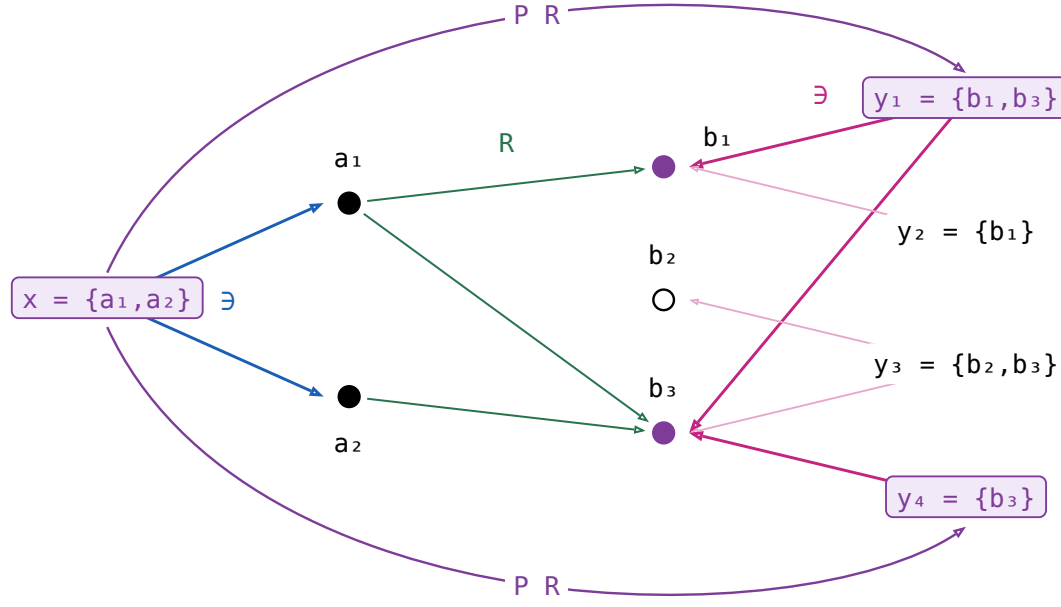
Dividing by $\exists$ turns that into a relation between **lists**: every element of y must be one of the filled dots.



$$(\exists R)/\exists : [A] \rightarrow [B] \quad x ((\exists R)/\exists) y \iff \forall b \in y. \exists a \in x. a R b$$

y_3 is rejected: it names b_2 , which x does not reach

The other half of the meet is that clause with the lists swapped and R turned round: every element of x must reach y . Both together, and two more lists to separate them:



$$((\exists R^\circ)/\exists)^\circ : [A] \rightarrow [B] \quad x ((\exists R^\circ)/\exists)^\circ y \iff \forall a \in x. \exists b \in y. a R b$$

y_2 is rejected too: a_2 's only image is b_3 , which y_2 does not name

In words, if $x (P R) y$ then every element of x is related by R to some element of y , and conversely — the two clauses of n , one per direction.

Neither $\exists$ nor $P R$ is a map. $\exists$ sends a list to **each** of its elements; $P R$ sends x to **every** y meeting the two clauses, y_1 and y_4 both. Nor is it entire: an $a \in x$ with no image at all leaves x with no partner whatever. Turning a relation into a map is Λ 's job (§11), and on lists that map is $\Lambda(\exists R)$ — of the four it accepts y_1 only.

Not a symmetric division. $\frac{\exists R}{\exists}$ is $\Lambda(\exists R)$, and in $\mathbf{Rel}$ it reads $x \Lambda(\exists R) y \iff y = \{b \mid \exists a \in x. a R b\}$: in words, y is **exactly** what x reaches, where $P R$ asks only that each side cover the other. Nor can it be repaired: in that fraction's second half $(\exists/(\exists R))^\circ$ the R sits in a denominator, so the operation is antitone there, and a relator has to be monotone. $P R$ keeps the first half and takes the second half at R° , which puts R back in a numerator. The fraction returns exactly where the two halves agree — at 1 , and at a map.

law	the reading
$X \sqsubseteq P R \iff X \exists \sqsubseteq \exists R$ and $X^\circ \exists \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = (P R)^\circ$, and $R \sqsubseteq S \implies P R \sqsubseteq P S$.
$P 1 = \frac{\exists}{\exists} = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$P f = \Lambda(\exists f) = \frac{\exists f}{\exists}$, for f a map	In $\mathbf{Rel}$, $x (P f) y \iff y = \{f a \mid a \in x\}$. The half at f° says every $a \in x$ has its $f a$ on y ; f has just the one image per a , so that already says y contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R S) = (P R)(P S)$	$\sqsubseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R S)$.

§12. Catamorphism

F a relator with an **initial algebra** $\alpha : F T \rightarrow T$ among the maps. For a relational algebra $R : F B \rightarrow B$, the **catamorphism** $\langle R \rangle : T \rightarrow B$ is the unique arrow with $\alpha \langle R \rangle = (F \langle R \rangle) R$.

$$\begin{array}{ccc}
 F T & \xrightarrow{F X} & F B \\
 \alpha \downarrow & & \downarrow R \\
 T & \xrightarrow{X} & B
 \end{array}$$

With $X = \langle R \rangle$ the square commutes strictly: unlike the product's triangles, there is no slack here, so no $\sqsubseteq$ appears anywhere in it.

The three **fusion** rows rewrite $\langle R \rangle S$ through a second algebra $Q : F C \rightarrow C$ along an arrow $S : B \rightarrow C$. They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest of the table needs only the initial algebra.

name	law	what it says
the defining equation	$X = \langle R \rangle \iff \alpha X = (F X) R$	Folding a value that α has just built is the same as folding its parts and combining them with R . Nothing else has that property.
Lambek	$\alpha^\circ = \alpha^{-1}$, the initial algebra is an isomorphism	A constructor can be undone — α° takes a value apart into the parts it was built from.
Eilenberg–Wright	$\Lambda \langle R \rangle = \langle \Lambda((F \exists) R) \rangle$	A relational fold is a deterministic fold of SETS: Λ pushes the nondeterminism into the power-object, where the fold is a map again.
Eilenberg–Wright	$\langle R \rangle = \langle \Lambda((F \exists) R) \rangle \exists$	The same fact read back — fold deterministically into a set, then take a member of it.
fusion	$\langle Q \rangle \sqsubseteq \langle R \rangle S \iff (F S) Q \sqsubseteq R S$	Half of fusion: an inclusion between the two algebras is inherited by the folds.
fusion	$\langle R \rangle S \sqsubseteq \langle Q \rangle \iff R S \sqsubseteq (F S) Q$	The other half, with both inclusions turned around.
fusion	$\langle R \rangle S = \langle Q \rangle \iff R S = (F S) Q$	Both halves at once: a fold followed by S collapses into a single fold, which is how an intermediate structure is got rid of.

The two Λ rows, drawn:

$$\begin{array}{ccccc}
 F T & \xrightarrow{F K} & F [B] & \xrightarrow{F \exists} & F B \\
 \alpha \downarrow & & \downarrow \Lambda((F \exists) R) & & \downarrow R \\
 T & \xrightarrow{K} & [B] & \xrightarrow{\exists} & B
 \end{array}$$

The left square is that catamorphism's own defining square, $K = \langle \Lambda((F \exists) R) \rangle$, and the right one is Λ 's cancellation, so the outer rectangle says $K \exists$ satisfies the defining equation of $\langle R \rangle$ — and uniqueness finishes it.

§13. Type functor

F a **binary** relator: $F(R, S)$ is its action on a pair, and $F X$ abbreviates $F(1, X)$, the F of the catamorphism section. For every object A the initial algebra is $\alpha : F(A, T A) \rightarrow T A$, among the maps. The **type functor** T acts on an arrow $R : A \rightarrow B$ by

$$T R = (F(R, 1) \alpha) : T A \rightarrow T B$$

F is the **base functor** — one layer of the structure, acting on the recursive position — while T is the datatype itself, acting on the parameter. For cons-lists, `list R = ([nil, (R @ 1) cons])`, which is `map R`.

$$\begin{array}{ccccc}
 & & F(1, T R) & & F(R, 1) \\
 F(A, T A) & \xrightarrow{\quad\quad\quad} & F(A, T B) & \xrightarrow{\quad\quad\quad} & F(B, T B) \\
 \downarrow \alpha & & & & \downarrow \alpha \\
 T A & \xrightarrow{\quad\quad\quad T R \quad\quad\quad} & T B & &
 \end{array}$$

$T R$ is the unique arrow making it commute; there is no $\sqsubseteq$ in it.

name	law	what it says
the defining equation	$T R = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way; that is <code>map R</code> .
functor	$T 1 = 1$ and $(T R)(T S) = T (R S)$	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	$(T R) (Q) = (F(R, 1) Q)$	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of α	$\alpha (T R) = F(R, T R) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G R = F(R, T R)$ to T .
type relator	$(T R)^\circ = T (R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion row above applied to $T R$'s own defining algebra, whose side condition holds because F is a bifunctor — $F(R, 1) F(1, (Q)) = F(R, (Q)) = F(1, (Q)) F(R, 1)$ — so it inherits that row's local-completeness requirement.

$$\begin{array}{ccc}
 F(A, T A) & \xrightarrow{\quad F(R, T R) \quad} & F(B, T B) \\
 \downarrow \alpha & & \downarrow \alpha \\
 T A & \xrightarrow{\quad T R \quad} & T B
 \end{array}$$

It commutes strictly: it is the naturality square of α .